SOFTWARES REQUIRED:

1. STM32CubeMX
2. STM32CubeIDE
3. TouchGFX

Installing TOUCHGFX and Adding it in STM32CUBEIDE:

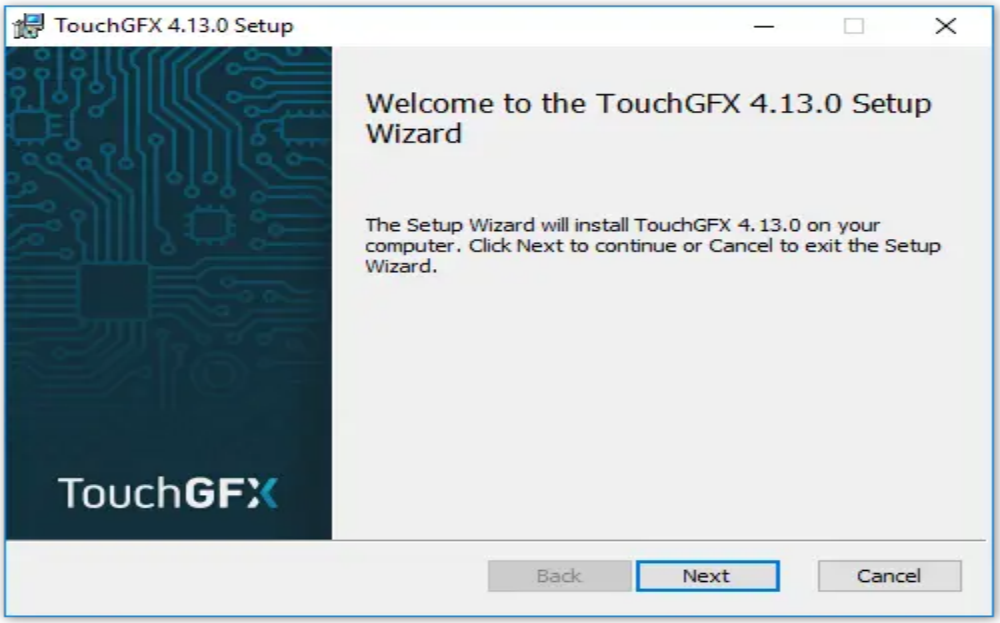
you can download X-CUBE-TOUCHGFX directly from the ST.com official website to anywhere on your hard drive and extract it.

Inside the unpacked/extracted X-CUBE-TOUCHGFX distribution you will find the TouchGFX .msi installer in the following path:

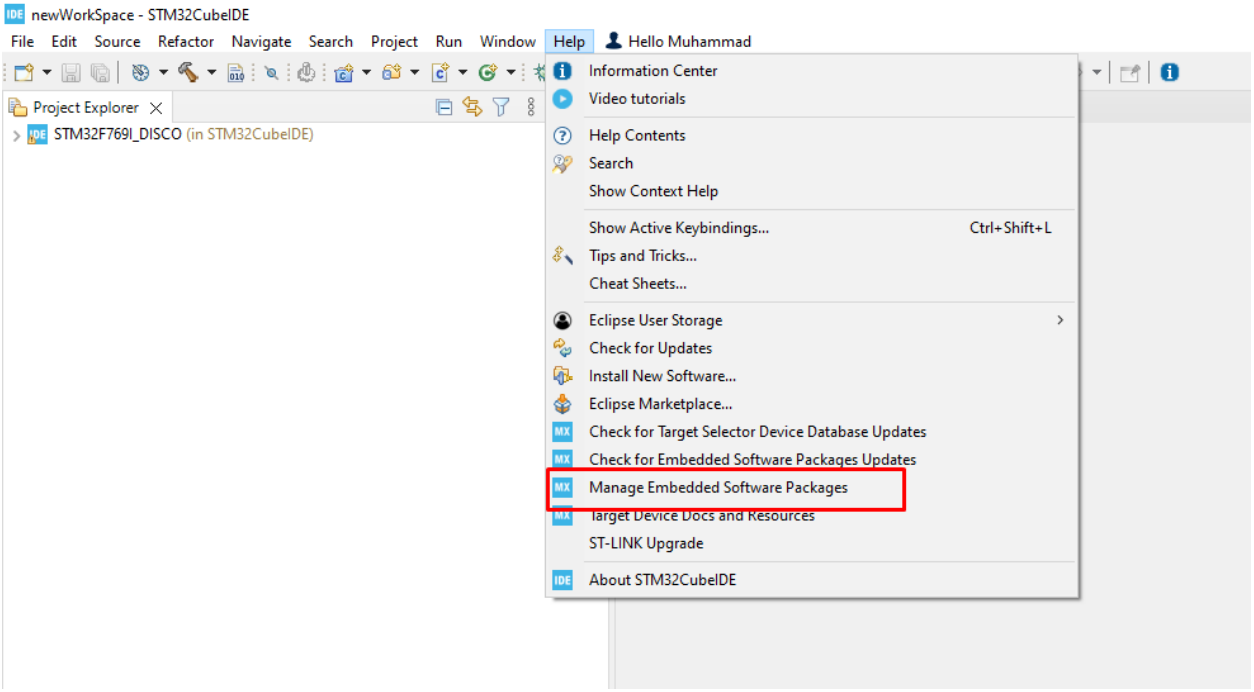
```
Unset

Utilities\PC_Software\TouchGFXDesigner
```

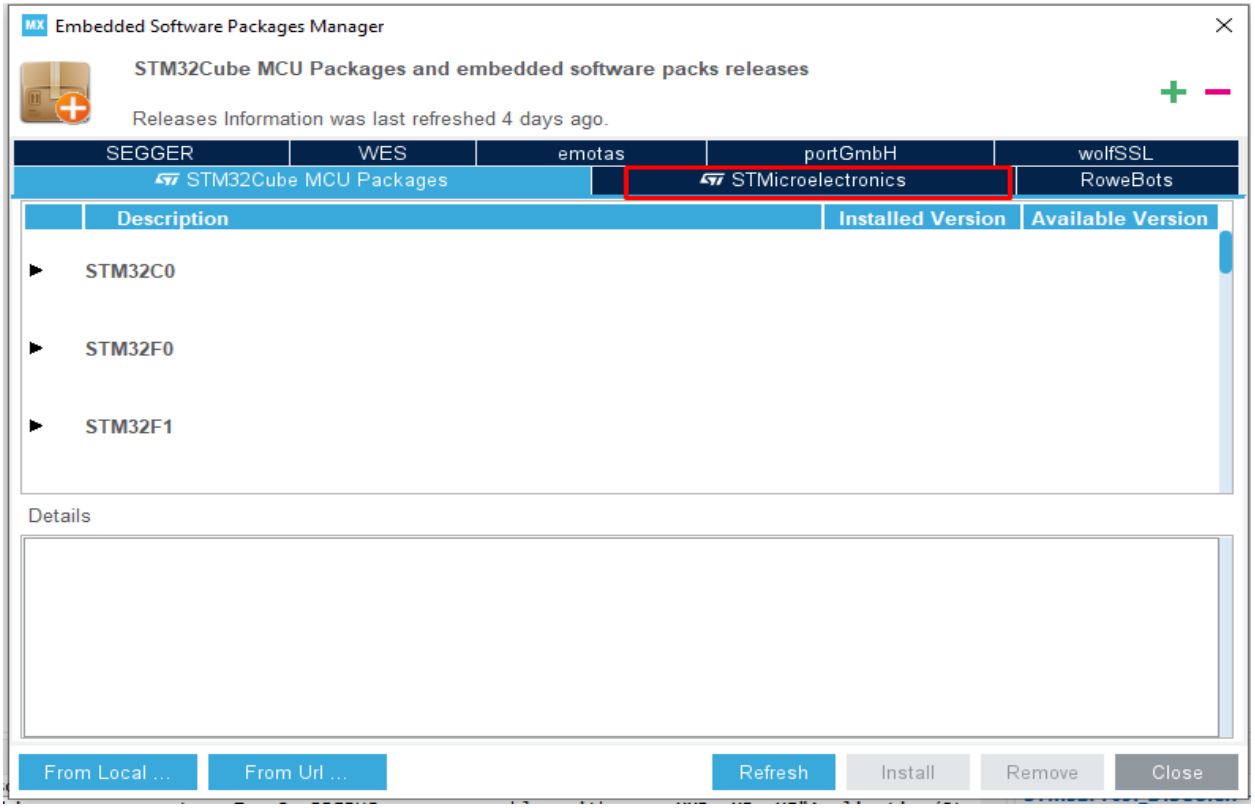
Double-clicking the .msi file will bring up the installer. Follow the instructions to complete the installation process.



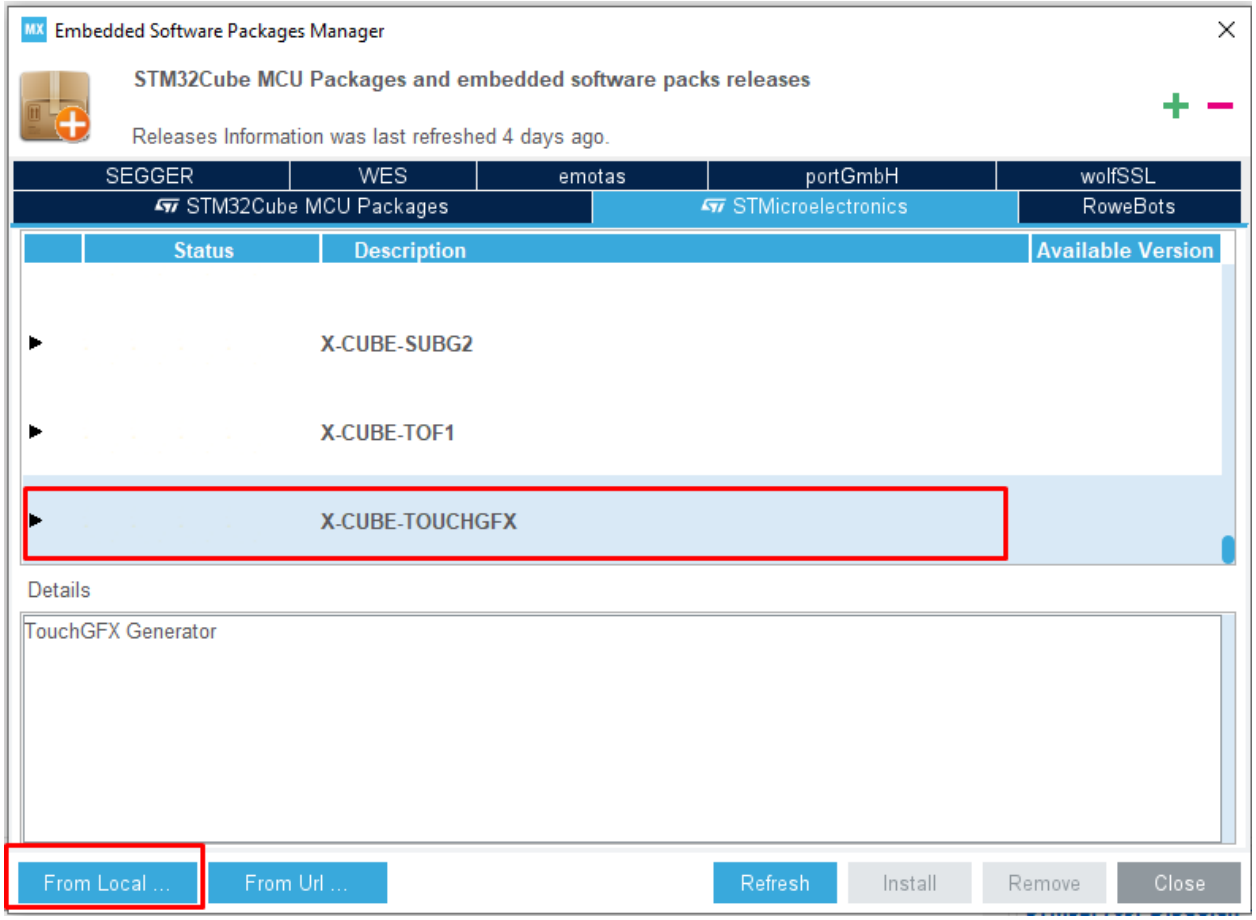
After Installation, open the project in STM32CUBEIDE and from the top click on Help>Manage Embedded Software Packaged



A window will open, head over to ST Microelectronics tab on that window,



Scroll down until you see X-CUBE-TOUCHGFX and click on “From Local” below and select the file that you have downloaded from ST website for touchgfx software.

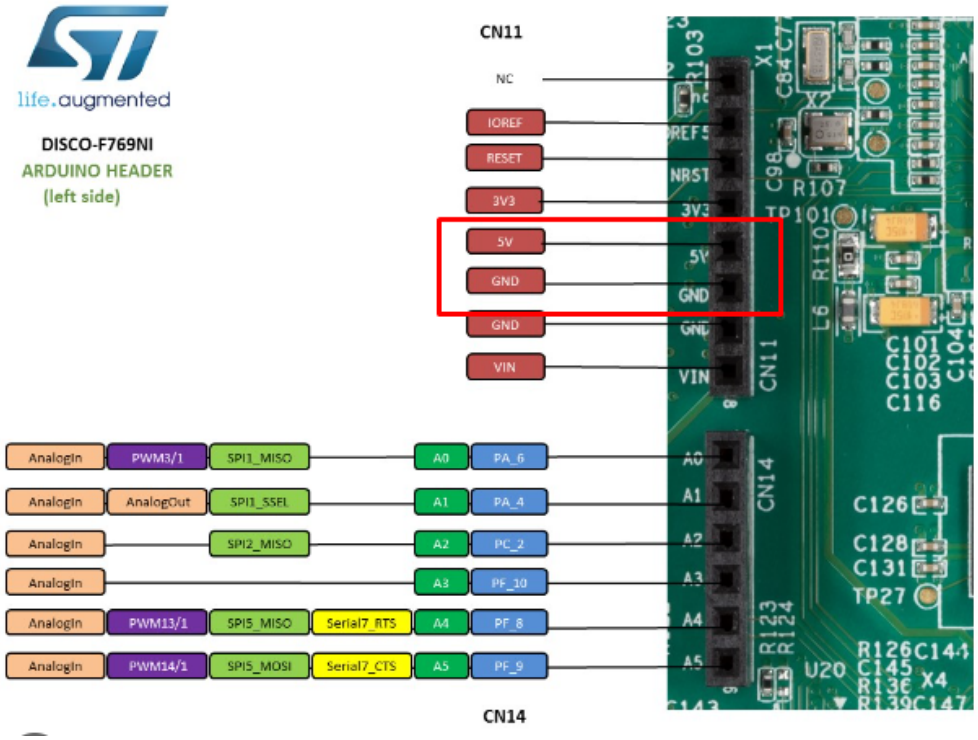


Wait for it to be finished then close the package manager.

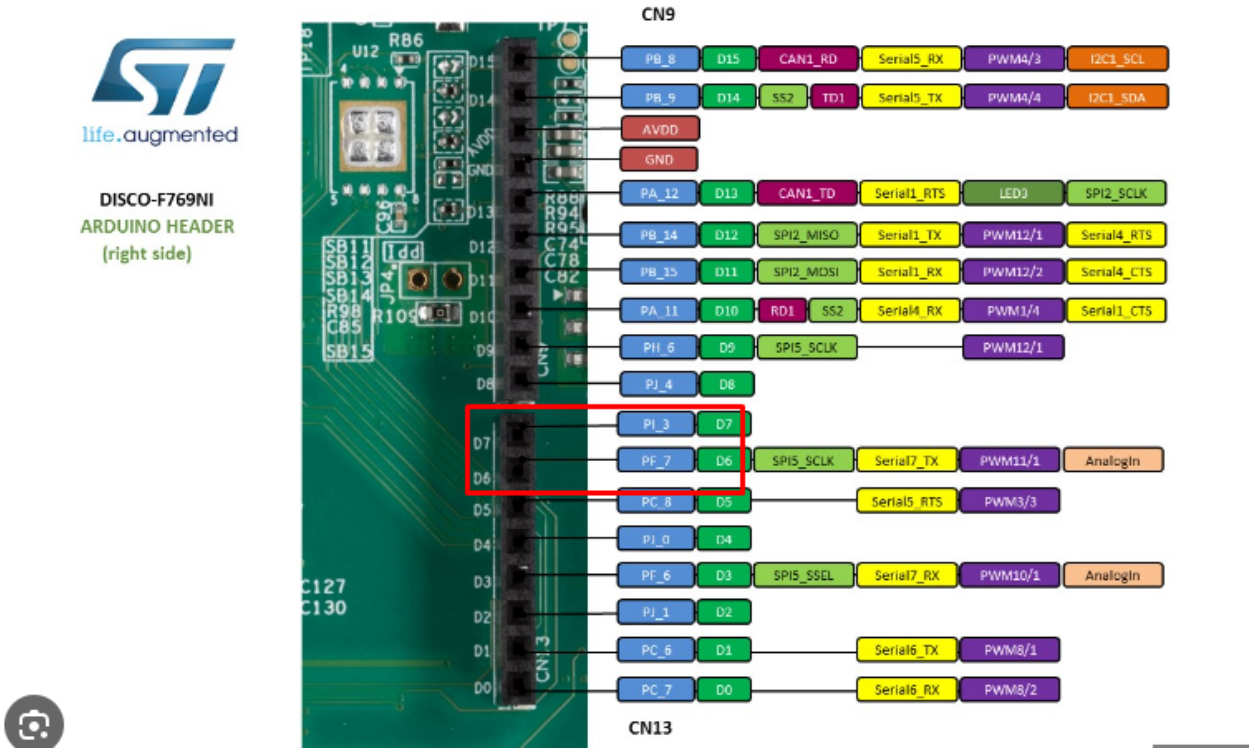
IMPORTANT: right click on the project name in the Tree View in STM32CubeIDE, select "Properties". Then in the "C/C++ Build " tab go to "Settings" and check under the MCU G++ Linker that you have the correct library path. For example, in my case my linker path is C:\TouchGFXProjects\guiApp\Middlewares\ST\touchgfx\lib\core\cortex_m7\gcc Similarly, find where your Middlewares\ST\touchgfx\lib\core\cortex_m7\gcc is located and update it in the linker path.

CONNECTIONS OF THE SENSOR WITH THE BOARD:

1. Connect the VCC of the sensor with the 5V pin of the board.
2. Connect the GND of the sensor with the GND pin of the board.



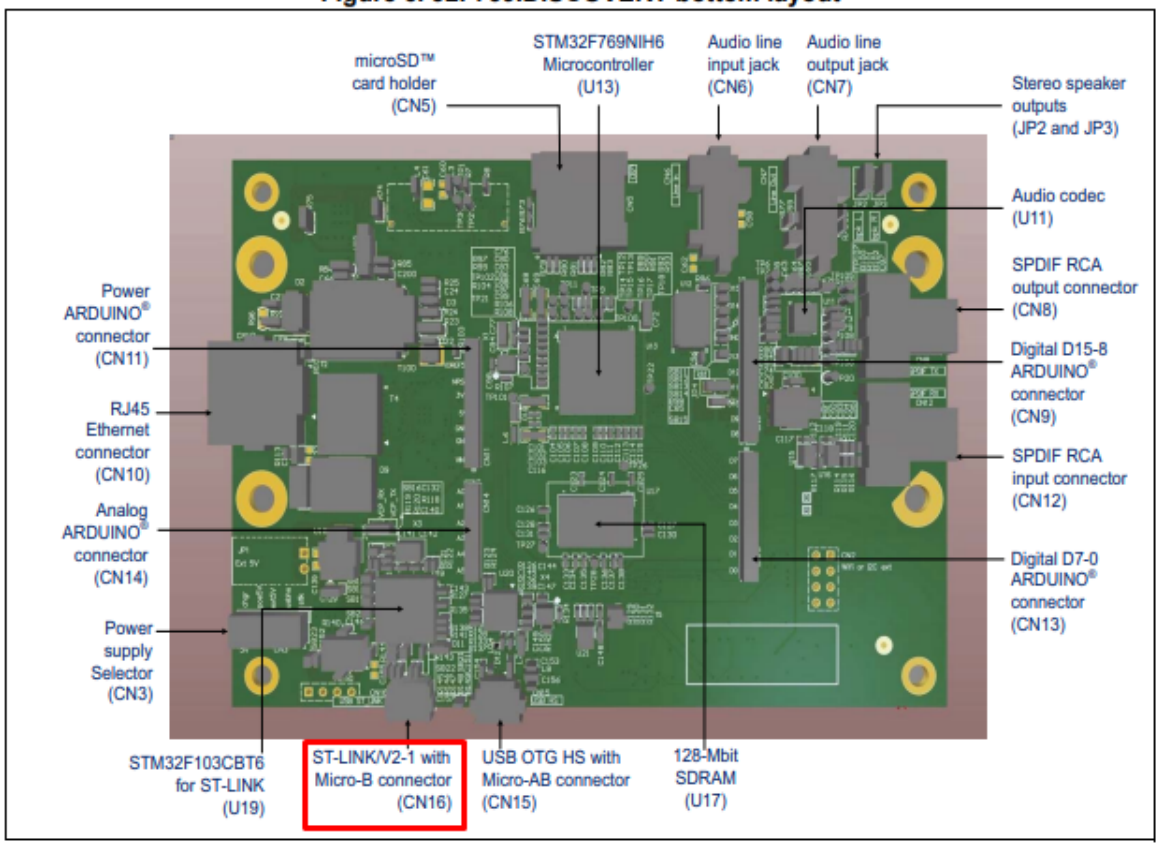
3. Connect the ECHO pin of the sensor with the D6 pin of the board.
4. Connect the TRIG pin of the sensor with the D7 pin of the board.



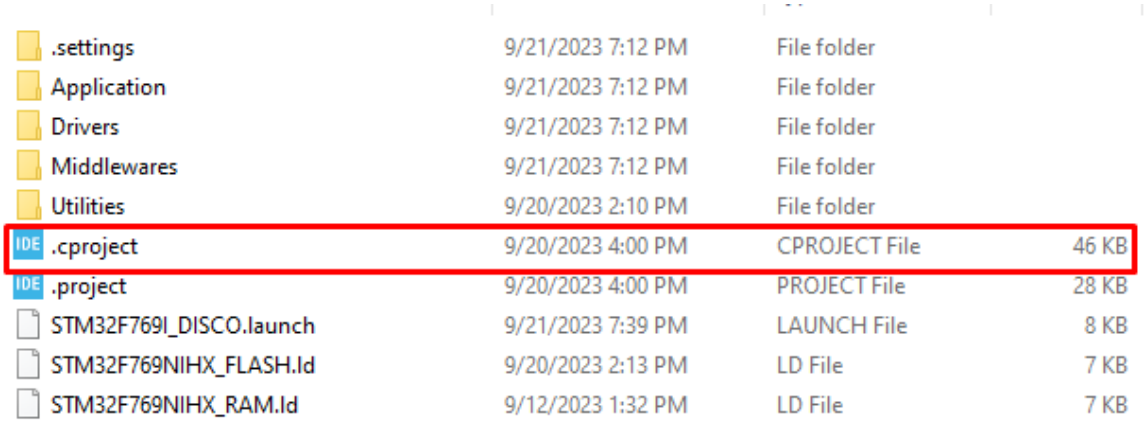
COMPILING & FLASHING THE FIRMWARE:

1. Connect the board with the PC through CN16 of the board.

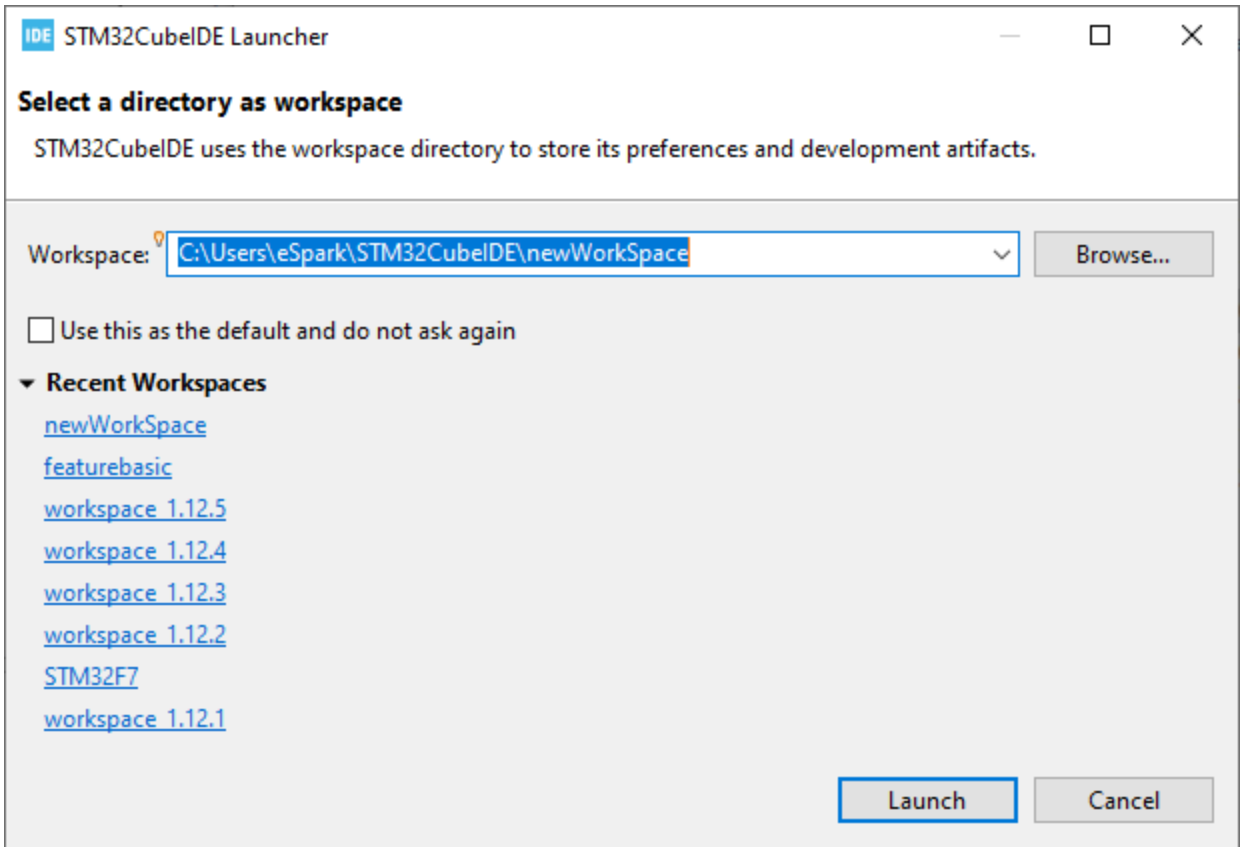
Figure 6. 32F769IDISCOVERY bottom layout



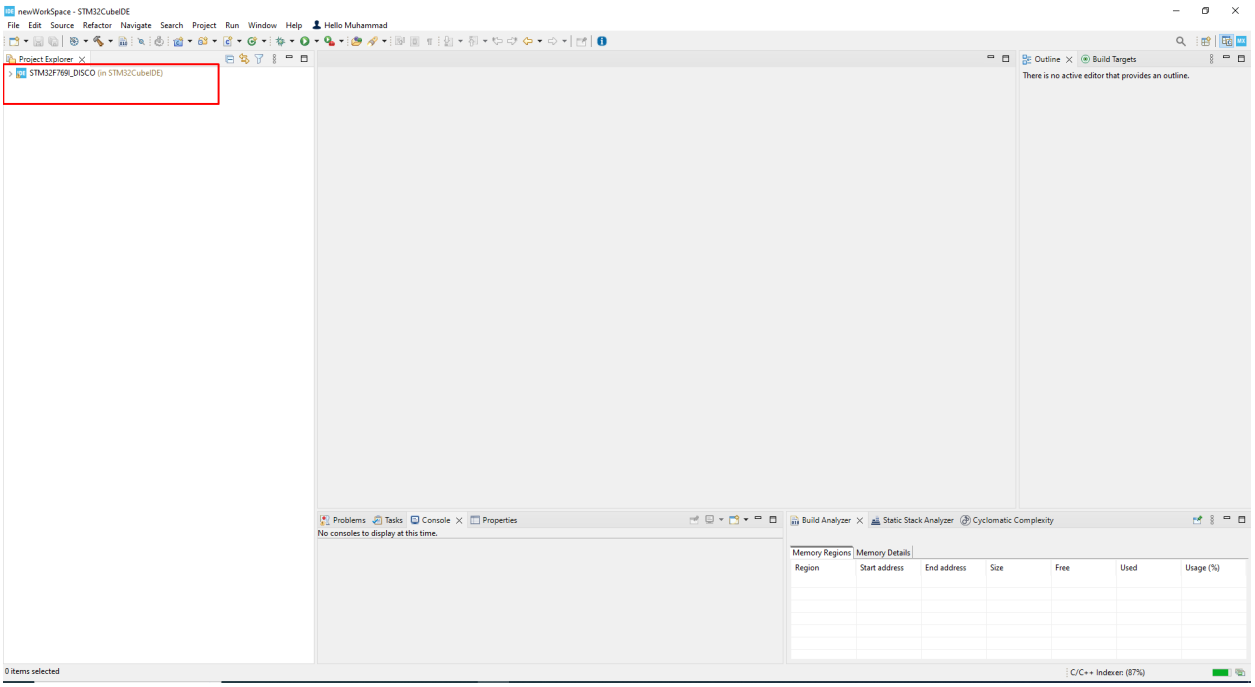
2. Open the .cproject file from the project directory.



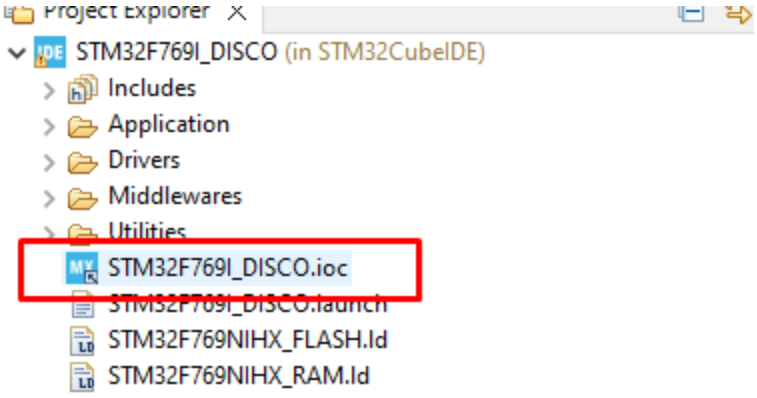
3. Create a new workspace or select an existing workspace in STM32CubeIDE



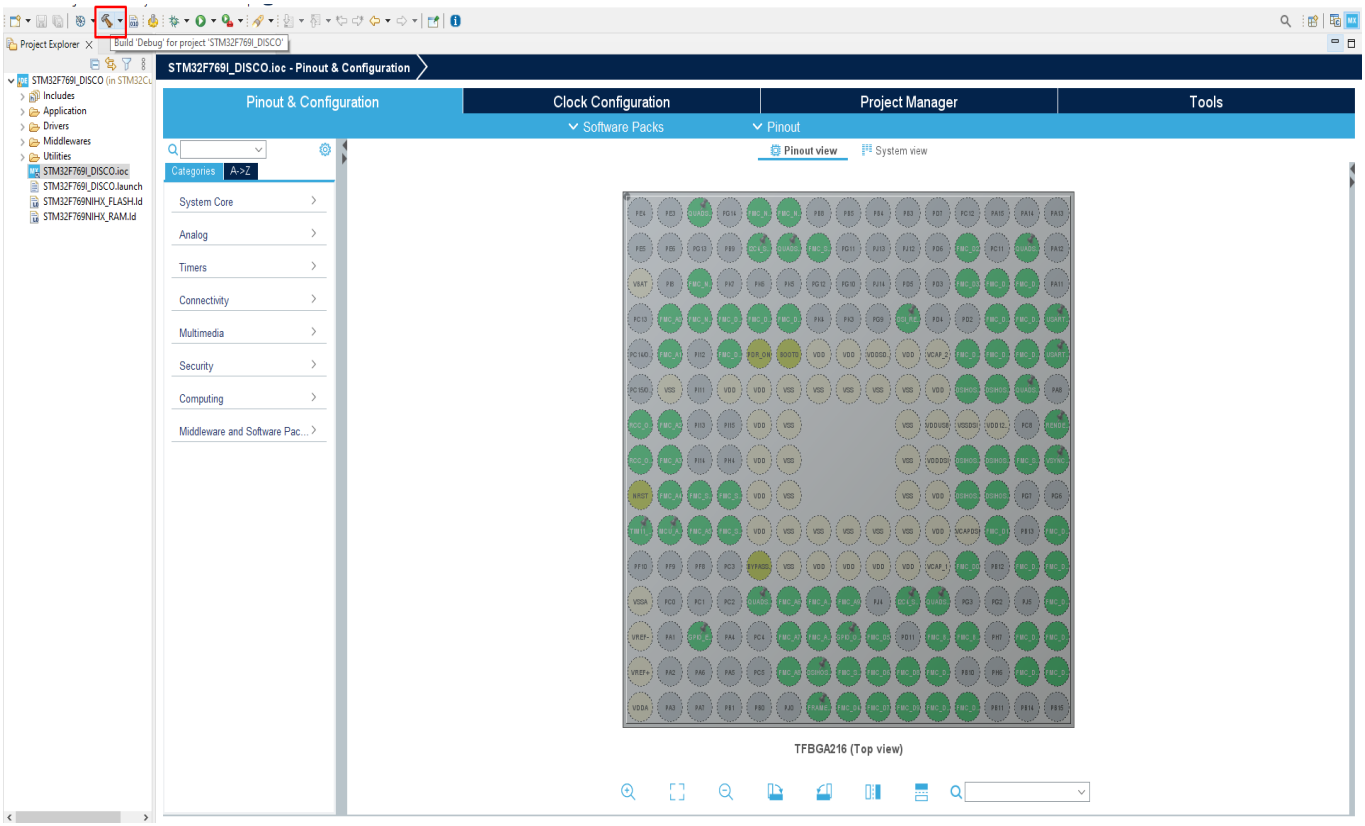
4. On the left, there is a project explorer pane where you can see the projects of the current workspace. The new project that you have opened must appear in that pane.



5. Open the dropdown menu of the project and double click on the “STM32F769I_DISCO.ioc” file.



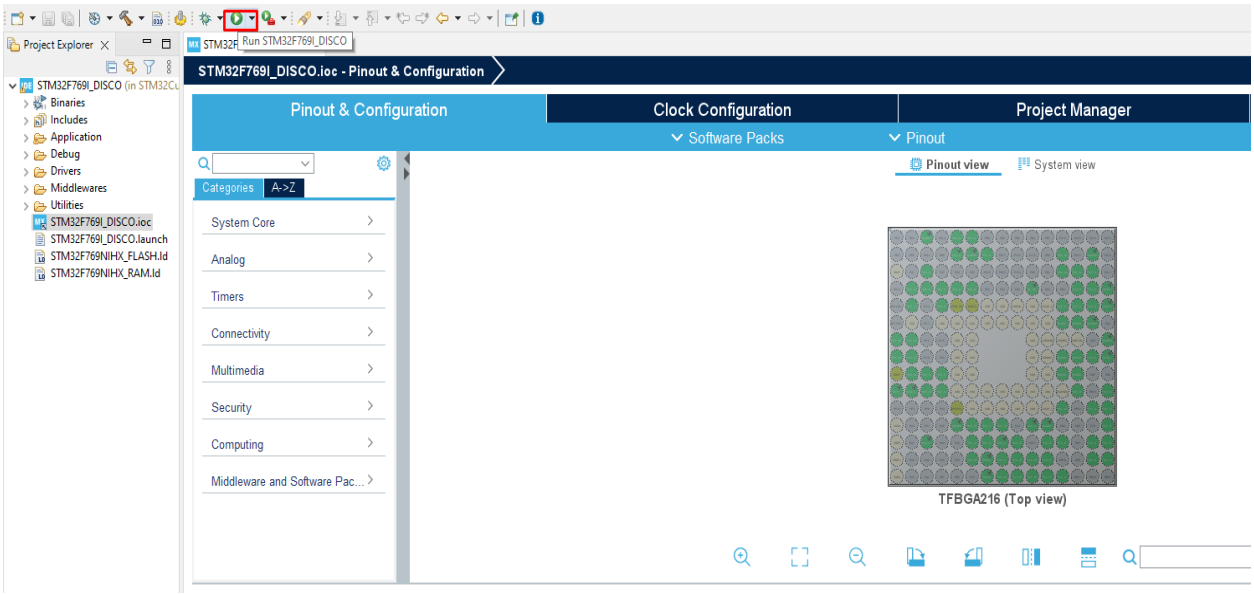
6. Click on the hammer icon on the top to compile the project.



7. Check the Console window to see if the project has been compiled.



8. Once the project has been compiled, click on the run icon on the top pane to run the project.



9. Check the Console again to see if the download has been verified successfully.



10. The firmware has been flashed now press the blue button on the board to capture the readings of the sensor.

